

A conditional 2^{-59} exponent saving for integer multiplication

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Abstract

A weighted paired-block circuit reduces the finite bit network, while two refinements of the downstream accounting permit a larger transform dimension: the guard bound uses the actual early stopping depth, and the Gaussian width is chosen to meet its stated interface directly. At ground size $h = 50$, these changes give the conditional saving $\kappa = 2^{-59}$, with a strictly positive exact margin. This is a factor $128/13$ above the preceding $13 \cdot 2^{-66}$ witness. The underlying multiplication theorem remains an assumption.

1 Scope

The source patch is against OpenAI's *Integer multiplication below $n \log n$* , pinned at

`adc7f1241b42e322a6451854ab7e4b4c146bf78a`.

The computational model uses a fixed finite alphabet and a fixed number of one-dimensional tapes. We retain nonadjacent routing, frames from orthogonal complements, and auxiliary stage sharing. The ground size is now 50 for both finite networks. The complex motif uses its original family; only the bit motif receives the new sum circuit. No practical runtime or unrestricted optimality claim is made.

1.1 A paired-block circuit for the bit correction

Use $h = 50$, $v = \binom{h}{3} = 19600$, $N = v^3$, and $m = h^3 = 125000$. The complex network above is retained at this ground size. For the bit network, we replace the intersection-one side correction by a smaller circuit. Fix a common point i . On the other $n = h - 1 = 49$ points, inputs x_{ab} represent source triples $\{i, a, b\}$. The required partial output is

$$z_{cd} = \sum_{\{a,b\} \cap \{c,d\} = \emptyset} x_{ab}.$$

All scalar additions are in $\mathbb{F}_2$, but the computation below always adds disjoint formal input supports.

It is useful to solve a weighted version. Give a graph an edge value e_{uv} and a vertex value w_u , with pairwise disjoint supports. Return its total value and the values left after deleting each one or two vertices, including all their incident edges. At the root, take $w_u = 0$ and $e_{uv} = x_{uv}$. Group the ordered vertices consecutively into pairs, with a final singleton when necessary. A block A becomes a vertex of a smaller graph with weight

$$\tilde{w}_A = \sum_{u \in A} w_u + \sum_{\{u,v\} \subset A} e_{uv}, \quad \tilde{e}_{AB} = \sum_{u \in A, v \in B} e_{uv}.$$

*Prepared with assistance from OpenAI Codex. This is not independent review or endorsement by OpenAI.

Recursively compute the smaller graph's total, its value O_A after deleting block A , and its value F_{AB} after deleting both A, B .

For $a \in A$, put

$$C_a = \sum_{u \in A \setminus \{a\}} w_u, \quad E_{a,B} = \sum_{u \in A \setminus \{a\}, v \in B} e_{uv}, \quad H_{a;B} = C_a + \sum_{C \neq A,B} E_{a,C}.$$

Compute all $H_{a;B}$ by the leave-one-out prefix/suffix routine on the vector $(C_a, (E_{a,C})_{C \neq A})$, requesting only the outputs that omit one of the E terms. The total of this vector is denoted H_a . If a, b are in different blocks A, B , their exclusion output is

$$F_{AB} + H_{a;B} + H_{b;A} + \sum_{u \in A \setminus \{a\}, v \in B \setminus \{b\}} e_{uv}.$$

For two vertices in one block it is O_A ; the single exclusion of a is $O_A + H_a$. These are disjoint partitions of the surviving weighted graph. This proves the recursion by induction. The weights created by internal edges remain inside the recursion, avoiding a separate exclusion circuit for them.

Use direct sums for at most four vertices. Compute other totals by balanced binary addition. In the four-term formula, first add $F_{AB} + H_{a;B}$, then add the other two terms, then combine those two results. Intern equal formal supports and remove unused nodes. At $n = 49$, the exact circuit has 9813 additions and 1176 designated outputs. All 1271256 nonzero coefficients, and every required zero, are checked by comparing all output supports with their definitions.

Identify equal inputs and equal sums across the h common-point groups. A sum shared by groups $i \neq j$ consists entirely of triples $\{i, j, k\}$; its fixed pair and variable vertices identify it exactly. Keep the first encountered decomposition, in common-point and then local-node order. Retain all $q = h \binom{h-1}{2} = 58800$ partial outputs, injecting their sum into each physical target. Since every neighbor pair has a unique common point, this remains exactly the intersection-one matrix. The global graph has

$$c = 450394, \quad q = 58800, \quad c + q = 509194.$$

Here 40256 additions have been merged from the separate local graphs.

1.2 Reversible roles and frames in both directions

For every node, regard its later uses and designated outputs as outgoing edges. At an input node, allocate one role per outgoing edge and copy a pivot into the others. At an addition node, add the second incoming role into the first, reuse the first as one outgoing pivot, and copy it into fresh roles for the other outgoing edges. The other input role retires. These are invertible pointwise gates. There are $2c + q$ outgoing uses and c pivot identifications, so $R = c + q = 509194$ roles suffice. Roles are compiled from the merged graph; no simultaneously occupied physical slots are identified.

Let L be the product of these gates, V the source copy, J the injection of the designated partial outputs, and G, R_0 the original central operations. The chronological schedule is

$$L, J, L^{-1}, R_0, V, G, R_0, L, J, L^{-1}, G, V.$$

The two injections give $JLz + JL(z + Vx) = JLVx$ for arbitrary initial scratch z . The cleanup restores z and all central scratch. Since $JLV + R_0G = I$, the invocation is $y \leftarrow y + x$. Reverse and invert the schedule for stage 2. Forward, inverse, forward invocations therefore exchange the two data banks.

Use the original rational form $I - J/9$ on $F = \mathbb{Q}^{50}$. At a stage, write $A = B \perp P$ as before, suppressing the future line factor Q . Set $D_0 = B \otimes F$, $D_1 = A \otimes F$, and $D_U = (B \otimes F) \perp (P \otimes U)$. Let U_z be the span of the source indicators reaching node z . Every retained node is a sum from an original common-point group, so all its triples contain some i . For a vector u in that span,

$$\sum_j u_j = 3u_i, \quad \langle u, u \rangle = \sum_{j \neq i} u_j^2 > 0 \quad (u \neq 0).$$

Thus U_z is nondegenerate. Along every graph edge $z \rightarrow w$, $U_z \subset U_w$. Forward middle gates use D_{U_z} ; the physical roles follow paths, so these frames increase. A designated output for target T has $U_z \subset t_T^\perp$, because every contributing triple is a neighbor.

For the reversed middle computation use $D_{U_z^\perp}$. The ambient form is nondegenerate, so all these complements are nondegenerate. Reversing an edge reverses inclusion: $U_w^\perp \subset U_z^\perp$. The original output begins at the physical X line $\langle t_T \rangle \subset U_z^\perp$; an original input ends at exactly its physical Y complement $t_S^\perp$. No positivity assumption on the reachable-target spans is needed. At every gate all incident roles receive the same frame. Early mixers use D_0 , and late cleanup uses D_1 . Newly active roles enter from D_0 and retired roles grow to D_1 . Tensoring by Q gives the original data-stage boundaries in both orientations. The only decreases remain the central returns, each of dimension h .

Share the complete auxiliary bank of a stage-1 invocation (A, B) with stage 3 at $(B, \pi(A))$, matching local role indices. Here π is a bijection of triples with $|T \cap \pi(T)| = 1$: pair consecutive ground points; for a triple with a full pair and singleton, keep the singleton and cycle the full pair among pairs not containing it; otherwise keep the point in the least indexed pair and flip the other two to their partners. Each rule is bijective on its class. All invocations restore arbitrary scratch, and the joining labels are

$$E = F \otimes \langle t_A \rangle \otimes \langle t_B \rangle, \quad H = \langle t_B \otimes t_{\pi(A)} \rangle^\perp \otimes F.$$

They are nondegenerate and $E \subset H$ by the chosen orthogonality. Each join removes one auxiliary role and exactly m of edge rank.

Proposition 1 (Paired-block bit interface). *The new bit network restores arbitrary scratch, exchanges the data banks, and satisfies the original all-role endpoint identity with*

$$\begin{aligned} W_b^{\text{pair}} &= 2N + 2v^2(R + h) = 406321422080000, \\ s_b^{\text{pair}} &= W_b^{\text{pair}}m - N + 6v^2h^2 = 50790175992864000000, \\ W_b^{\text{pair}}m - s_b^{\text{pair}} &= 1767136000000 > 0. \end{aligned}$$

Proof. The scalar schedule and both frame assignments were verified above. The total decreasing dimension is $3v^2h^2$. Telescoping gives total absolute label change $W_b^{\text{pair}}m - 2N + 6v^2h^2$; the unchanged negative source projections add N rational rank. Data endpoints are unchanged and every surviving auxiliary role has endpoints zero and I_m , with its scalar value restored. The original common-frame identity and rational-shear transfer apply. All circuits, rational bases, and gates are fixed finite data. $\square$

1.3 Explicit exponents at ground size fifty

Take $a = 296/10^{11}$, $a_c = 1/10^{11}$, and $L_0 = 11737/1000$. The exact deficits satisfy

$$\eta_b^{\text{pair}} = \frac{23}{661055000} > aL_0, \quad \eta_c = \frac{239}{1202215191250} > a_cL_0.$$

For $1 \leq x \leq 2$, put $z = (x-1)/(x+1)$, $S(x) = 2 \sum_{j=0}^{23} z^{2j+1}/(2j+1)$, and $E_0(x) = 2z^{49}/(49(1-z^2))$. Then $S(x) \leq \log x \leq S(x) + E_0(x)$, and exact rational comparison gives

$$\log m \leq 16(S(2) + E_0(2)) + S(m/2^{16}) + E_0(m/2^{16}) < L_0.$$

The inequality $e^{-x} > 1 - x$ therefore proves the required strict bounds with

$$\tau = 1 - \frac{296}{10^{11}}, \quad \sigma = 1 - \frac{1}{10^{11}}. \quad (1)$$

The complex network itself is unchanged apart from the ground size $h = 50$; its construction, binary residual bases, and scalar corrections remain those proved above.

2 The guard refinement

2.1 A guard bound using the actual stopping depth

The complex network is still evaluated exactly until a complete normalized butterfly layer is finished. Write $W = W_c$ and $s = s_c$ for that network, and retain

$$E = 64(W + m + 1)^3, \quad B = s + E, \quad C_0 = 128mB^2.$$

At $h = 50$ the exact values are

$$m = 125000, \quad W = 369474390296480000, \quad s = 46184298777878576000000,$$

so $m \geq 3$ and $2 \leq s < m^5$ by integer comparison. We will use $C_1 = 2$ and $\Delta = \lceil C_0 d^2 \rceil$ for every rational $9/10 \leq \beta < 1$.

Measure dependency depth by complex additions, subtractions, halvings, sign changes, and multiplication by i . A chain follows an operand into a later result. Concatenating every scalar gate and recursive edge call is a valid upper bound, even though most do not belong to one dependency chain. In the retained complex motif each wire touches at most twelve scalar gates, so there are at most $12W$ gates. A gate output sums at most W terms, with at most a halving and an addition per term. Evaluating all outputs from saved inputs uses at most $2W^2$ operations per gate. Inverse-child corrections cost at most $4s$, and source, sink, and bank corrections at most $4W + 4$. Thus their total is bounded by $24W^3 + 4s + 4W + 4 < E$. The new bit motif performs exact permutations of coefficient encodings; it does not introduce coefficient arithmetic into this depth count.

The individual C -kernel formula uses at most eight elementary operations. Consequently the same conservative recurrence remains valid:

$$A(e) \leq sA(e/m) + E \quad \text{at an internal node}, \quad A(e) \leq 8e \quad (e < d^\beta).$$

For a root $e \leq d$, the actual number j of internal levels is at most $(1 - \beta) \log_m d + 1$. If $j > 0$, this follows from the last internal size $e/m^{j-1} \geq d^\beta$; for $j = 0$ it is immediate. Hence

$$s^j \leq s d^{5(1-\beta)}, \quad A(e) \leq 8d^\beta s^j + E \sum_{i < j} s^i \leq s(8 + E) d^{5-4\beta} \leq 9B^2 d^{7/5}.$$

Here $s \geq 2$, $d \geq 1$, $\beta \geq 9/10$, and $s(8 + E) \leq 9B^2$; the estimate also covers a root that is already a leaf. Keeping E additive is essential: it is a fixed cost at a node, not another branching factor.

The base- m partition has at most $(m - 1)(1 + \lfloor \log_m d \rfloor)$ pieces. Since $m \geq 3$ and $\log d \leq \sqrt{d}$ for $d \geq 1$, this is at most $2m\sqrt{d}$. Concatenating their depths therefore costs at most $18mB^2 d^{19/10}$.

Preprocessing uses at most d individual kernels, costing $8d$. The two outer diagonals and the scalar b^D contribute at most $t + 9$ further units, where $t = \lfloor D/2 \rfloor$; a shift by t is charged t denominator bits. Together $8d + t + 9 \leq 18d$. Thus the total depth and shift charge are bounded by

$$18mB^2d^{19/10} + 18d \leq 36mB^2d^2 < C_0d^2.$$

As in the original induction, an input in $2^{-p}\mathbb{Z}[i]$ with modulus at most one has denominator exponent at most $p + \Delta$ and magnitude at most 2^Δ throughout. Reserving $p + \Delta$ fractional bits and $\Delta + 3$ integer and sign bits per real component therefore represents all values exactly. Address-swap routines restore complete encodings before coefficient arithmetic resumes and do not change this argument. If $2\epsilon < 1$, then $\Delta = o(p)$ and the full coefficient width remains $O(p)$.

3 A smaller Gaussian width

Keep $p = 6b$, $\eta = 1/(4d)$, and the same prime intervals. They still give $\theta_i = t_i/s_i - 1 > 1/(4d)$. Replace the earlier Gaussian width by

$$\alpha = \lceil (32db)^{1/4} \rceil, \quad \gamma = 2d\alpha^2.$$

Then $\alpha^4\theta_i > 8b > p$, precisely the strict hypothesis needed by the unchanged one-dimensional resampling interface. Also

$$\alpha \leq 2(32db)^{1/4}, \quad \gamma < 46d^{3/2}\sqrt{b}.$$

For $d = \Theta(p^\epsilon)$, we now have $\alpha = \Theta(p^{1/4+\epsilon/4})$ and $\gamma = O(p^{1/2+3\epsilon/2})$. Thus $\epsilon < 1/3$ implies $\gamma = o(p)$ and $2 \leq \alpha < \sqrt{p}$ eventually. The previous convenient condition $\epsilon < 1/12$ and its bound $\gamma \leq 28b^{2/3}$ are not needed.

The unchanged Gaussian line-map cost $O(dTp^{3/2+\delta}\alpha)$ now has normalized exponent $3/4 + \delta + 5\epsilon/4$. The other six assembly exponents are unchanged. Prime selection still requires $1 - 2\epsilon > 0$; the product of the selected primes remains between $T/2$ and T , with the same separation and coprimality guarantees. The new width satisfies every stated resampling hypothesis without changing that analytic lemma.

For the chosen $\epsilon = 199/1000$, the exponents of α and γ are respectively $1199/4000$ and $1597/2000$. The explicit cutoff $b \geq 2^{40}$ ensures $46b^{1597/2000} \leq b/4$ by integer comparison $184^{2000} < 2^{16120}$. Replacing the former 2^{24} cutoff by 2^{40} therefore preserves the source-scaling and final rounding bounds, together with the other existing eventual cutoffs. All grid, contraction, and error lemmas are retained with their original hypotheses verified.

4 Exact parameters and final margin

With $a = 296/10^{11}$ and $a_c = 1/10^{11}$ take

$$\begin{aligned} \tau &= 1 - a, & \sigma &= 1 - a_c, & \beta &= 999/1000, & \epsilon &= 199/1000, & \delta &= 1/10000, & C_1 &= 2, \\ c &= 999a/1000, & \lambda &= 1 - 1999a^2/2000, & \lambda' &= 1 - 999a^2/1000. \end{aligned}$$

Here $\tau(1 + c/\beta) = 1 - a^2 < \lambda < \lambda' < 1$; $\sigma < \lambda$ and $\sigma + \beta(1 - \sigma) < \lambda'$ hold by exact rational comparison. The stopping test is $e^{1000} < d^{999}$. The seven margins, with nonadjacent routing and the new Gaussian width, are

$$\begin{aligned} g_1 &= 1 - \epsilon(1 + c), & g_2 &= \epsilon c(1 - \tau), \\ g_3 &= \epsilon(1 - \lambda'), & g_4 &= (1 - \tau)(1 - \epsilon), \\ g_5 &= 1/4 - \delta - 5\epsilon/4, & g_6 &= 1 - \delta - \epsilon, & g_7 &= \epsilon. \end{aligned}$$

Exact arithmetic gives

$$G = \min_j g_j = g_2 = g_3 = \frac{272158569}{1562500000000000000000000} > 2^{-59}.$$

The ratio $G/2^{-59}$ is approximately 1.00408789; the displayed strict rational comparison, not this decimal, is the certificate. The fixed gap absorbs the remaining fixed powers of $\log p$.

The guard exponent is $2\epsilon = 199/500 < 1$, the prime-interval growth exponent is $301/500 > 0$, and $\ell = \Theta(p^{801/1000})$. We have $K = \Theta(p^{\epsilon c}) = o(\ell)$ and $K/\log p \rightarrow \infty$. The Gaussian margin is $23/20000 > 0$. Thus all setup, stopping, precision, layout, and absorption conditions in the patched proof hold, yielding

$$T(n) = O(n(\log n)^{1-2^{-59}})$$

under the retained upstream assumptions.

5 Reproducibility and boundary of verification

The reproducible artifacts are

- `certificates/paired-network.json`;
- `patches/h50-paired-59.patch`;
- `scripts/paired_exclusion_circuit.py`;
- `scripts/shared_point_circuit.py`;
- `scripts/paired_network.py`.

The checks compare every local and global output support exactly, verify both frame directions on every compiled role, and check small forward and reverse invocations on every input basis vector, including dirty scratch. Full small three-stage models check bank exchange and shared auxiliary restoration. Separate tests cover weighted recursion, stopping-depth bounds, guard-model selection, revised assembly constraints, and source-patch dependencies. These checks support the written proofs; they do not independently establish or formally verify the complete upstream algorithmic theorem.